APPAREL MANUFACTURING QMS v2.4.0

GARMENT MANUFACTURING QUALITY MANAGEMENT SYSTEM

Closed-Loop Non-Conformance (NC) & Corrective Action (CAPA) Platform

Complete Technical Specification, Enterprise Architecture, Database Schemas, RESTful API Documentation, Compliance Scoring Algorithms, and Standard Operating Procedures (SOP) for apparel factory shop floor defect rectification & SLA enforcement.

TARGET DOMAIN	SLA ENFORCEMENT	PROOF & AUDIT TRAIL
Textile & Garment Plants	**Strict 12h - 24h Timers**	**Closed-Loop Verification**
Cutting, Sewing, Finishing, Packing	Critical (12h) / Major (24h) SLAs	Before/After Visual Photographic Evidence

CORE ENGINEERING PILLARS

- **Role-Based Portals:** Dedicated security views for Plant Head/Admin, Floor Auditor, and Department Line Supervisors
- **9-Field CAPA Engine:** Standardized ISO-aligned root cause analysis, containment, preventive actions & verifiable criteria
- **Photographic Proof:** Pre-compressed mobile camera uploads embedded directly into offline-ready PDF reports
- **Real-Time Floor Radar:** Live employee attendance, worker task progress, and department compliance benchmarking
- **Weighted Scoring Engine:** Mathematical audit compliance algorithms with risk band classification (Critical, High, Medium, Low)
- **Offline Resilience:** Zero-data-loss architecture with resilient in-memory storage fallback and MongoDB persistence

DOCUMENT ID: QMS-SPEC-2026-V2.4
CLASSIFICATION: CONFIDENTIAL / INTERNAL USE ONLY
AUTHOR: DeepMind Advanced Agentic Systems & QMS Engineering Team
RELEASE: Production Release
DATE: October 2026
STATUS: Verified & Audit-Approved

DOCUMENT NAVIGATION

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Executive Summary

In modern apparel manufacturing, quality non-conformances (fabric holes, missing stitches, shade variation, oil stains, measurement skews) represent substantial financial write-offs and customer delivery penalties when caught late in the packing or shipping stages. Traditional paper-based quality registers and ad-hoc chat channels fail due to lack of traceability, delayed supervisor containment actions, missing visual evidence, and zero executive accountability.

The Garment Manufacturing Quality Management System (QMS) delivers an automated, closed-loop defect rectification platform. It bridges roving quality inspectors directly with production line supervisors under strict SLA countdowns (12 hours for Critical defects, 24 hours for Major defects). By demanding photographic evidence before and after rectification, and computing instant plant-wide compliance ratings, the system achieves 100% defect traceability from floor discovery through final auditor sign-off.

Key Business Objective

Reduce garment defect recurrence by 65%, cut rework cycle times from 72h to under 18h, and guarantee 100% compliance audit readiness for international apparel buyer standards (AQL 1.5/2.5).

SYSTEM CONTEXT

1. Executive Overview & Problem Statement

Analyzing shop-floor defects and the closed-loop digital solution

1.1 Shop-Floor Quality Realities

Textile assembly lines operate at fast throughputs (hundreds of pieces per hour per line). When a sewing operator misaligns a collar or an oil leak stains a bundle, every subsequent minute causes tens of defective units to be produced. The core operational bottlenecks addressed by this platform include:

Identified Bottleneck	Operational Impact & Factory Cost
Paper Register Latency	Auditors record defects on physical clipboards. Line supervisors only see defect notes hours later during shift changes, creating massive batches of defective garments.
Ambiguity in Containment	Notes such as "Fix stitching on batch 4" provide no precise visual context. Supervisors often perform partial fixes that fail re-inspection.
SLA Drift & Lack of Escalation	Without automated SLA timers, non-conformances linger in "Open" status for days with zero automated notification to plant management.
Inactivity & Floor Visibility Gaps	Plant managers lack real-time visibility into supervisor responsiveness, employee availability, or department workload.

1.2 The Closed-Loop Digital Solution

Garment QMS replaces paper registers with an immediate digital workflow. Floor auditors photograph defects via mobile devices; the system compresses images locally, extracts GPS/line metadata, assigns the complaint to the designated supervisor, and arms a real-time countdown timer. Once the supervisor completes root-cause analysis and uploads after-rectification photos, the ticket shifts to "Under Review". The auditor must physically verify the fix to either approve & close or reject for rework.

ISO 9001 / IATF Quality Alignment

The 9-field Corrective and Preventive Action (CAPA) model enforces structured 5-Why root cause isolation, immediate containment, and long-term recurrence prevention.

TECHNICAL ARCHITECTURE

2. System Architecture & Technology Stack

High-performance, offline-resilient, full-stack implementation

2.1 Architectural Topology

The application follows a decoupled three-tier architecture engineered for low-latency floor operations, mobile responsiveness, and high data integrity:

Architectural Tier	Technology Stack & Functional Role
Client Tier (Presentation)	React 18 SPA built with Vite 5. Tailored responsive dark/light dashboards, Tailwind CSS utility styling, Lucide icon system, and HTML5 Canvas image optimization engines.
Application Tier (API & Logic)	Node.js (v18+) and Express.js 4 REST API server. JWT authentication middleware, Multer multipart upload processor, SLA calculation engine, and weighted scoring utilities.
Persistence Tier (Data Store)	MongoDB document database via Mongoose ODM. Schemas enforce validation, timestamps, and embedded audit logs. Built-in resilient mock store ensures uninterrupted testing during database maintenance.
Reporting Tier (Document Export)	Client-side and server-side PDF compilation utilizing jsPDF and AutoTable. Embeds high-fidelity JPEG photographic proofs directly into standalone, printable certificates.

2.2 Technology Stack Matrix

Component	Package / Version	Engineering Purpose
Frontend Framework	React 18.3.1	Component-based UI with hooks, context state management, and error boundaries
Build Tool	Vite 5.4.14	Instant HMR, ES modules bundling, and optimized production chunking
Styling & Design	Tailwind CSS 3.4	Custom design tokens, slate-navy dark mode palette, and micro-animations
Backend Runtime	Node.js 18+ / Express 4.21	Non-blocking I/O, REST endpoints, JSON/multipart payload processing
Database & ODM	MongoDB & Mongoose 9.11	Document-oriented schemas, schema validators, indexing, and connection caching
Authentication	JWT & BcryptJS	Stateless Bearer token auth, salt-hashed passwords, and role authorization
File Processing	Multer 1.4 & Canvas	Streaming disk/memory uploads with client-side JPEG 800px dimension normalization
Document Generation	jsPDF 4.2 & AutoTable 5.0	Programmatic PDF generation with vector graphics, auto-pagination, and image embedding

CORE DOMAIN MODEL

3. Domain Model & Non-Conformance (NC) Lifecycle

Dual-axis state machine, SLA timing, and 9-field CAPA structure

3.1 Dual-Axis State Decoupling

A central innovation in Garment QMS is the decoupling of Workflow Status from SLA Deadline Condition. In primitive systems, ticket status often conflicts with temporal deadlines (e.g. marking a ticket "Overdue" overwrites that it is currently "Under Review"). Our architecture maintains two completely independent state axes:

Lifecycle Axis	Permitted State Transitions
Workflow Status Axis	Draft -> Open -> In Progress -> CAP Submitted -> Under Review -> Rejected/Rework -> Verified -> Closed
SLA Condition Axis	Open (Within SLA) | Due Soon (<4h Remaining) | Overdue (SLA Breached) | Closed (Resolved)

3.2 The 9-Field CAPA Specification

Every Non-Conformance ticket requires a structured 9-field Corrective and Preventive Action block prior to closure:

CAPA Field	Operational Standard & Verification Requirement
1. Immediate Correction	Action taken on the floor immediately to contain the bad pieces (e.g., quarantine bundle).
2. Root Cause Analysis	Underlying technical/procedural reason determined via 5-Why or Ishikawa Fishbone.
3. Corrective Action (CAP)	Permanent fix applied to eliminate the root cause on current production.
4. Preventive Action (PA)	Process/machine modification to prevent occurrence on future production orders.
5. Responsible Person	Specific employee or supervisor assigned ownership of the corrective measure.
6. Target Completion Date	Agreed timestamp deadline for completing rectification works.
7. Verification Method	Physical test or visual standard to be applied (e.g., pull-test, 100% visual inspection).
8. Verification Criteria	Quantifiable acceptance threshold (e.g., zero oil marks on 20 consecutive units).
9. Visual Photo Proof	High-resolution photographic evidence captured after rectification for side-by-side verification.

SECURITY & GOVERNANCE

4. Multi-Role Security & Permissions Matrix

Role-Based Access Control (RBAC) across factory personnel

4.1 Role Profiles

To guarantee segregation of duties, the platform partitions privileges into three dedicated operational roles:

Role Profile	Scope of Responsibility & Authority
Executive Admin / Plant Head	Full factory oversight. Manages user credentials, configures SLA thresholds, reviews factory-wide trend analytics, inspects department rankings, and deletes obsolete records.
Internal Auditor / Quality Inspector	Roving floor inspection. Initiates NC tickets, captures Before-Photos, defines severity/deadlines, inspects submitted CAP proofs, and holds sole authority to Verify or Reject tickets.
Line Supervisor / Department Lead	Production line owner. Views line-assigned tickets, claims department pool tasks, logs containment actions, executes 5-Why root cause analysis, and uploads After-Photos.

4.2 RBAC Permissions Matrix

System Operation / Action	Executive Admin	Internal Auditor	Line Supervisor
Log New Non-Conformance (NC)	YES	YES	NO
Upload Before-Photo (Defect Finding)	YES	YES	NO
Start Work / Mark In-Progress	YES	NO	YES
Submit CAPA & After-Photo Proof	YES	NO	YES
Verify & Approve Closure	YES	YES	NO
Reject for Rework & Reopen	YES	YES	NO
Manage User Accounts & Reset Passwords	YES	NO	NO
Configure Compliance Scoring Thresholds	YES	NO	NO
Download Full Audit PDF Dossiers	YES	YES	YES
Delete Non-Conformance Records	YES	NO	NO

DATA PERSISTENCE

5. Database Schemas & Data Modeling

Production MongoDB collection schemas and indexing policies

5.1 Core Collections

The database layer uses Mongoose ODM with strict schema validation, automated timestamps, and pre-save password hashing. The primary collections include:

Model / Schema	Collection	Field Specification & Attributes
User Collection	*users*	employeeId (unique), name, email (unique), passwordHash, role (ADMIN, AUDITOR, SUPERVISOR), department, designation, isActive, isPresent, lastLoginAt, lastLogoutAt.
Complaint Collection	*complaints*	complaintId (unique NC code), department, category, riskSeverity, status, deadlineTimestamp, beforePhoto, afterPhoto, assignedTo, cap (embedded 9-field object), auditTrail (array of status events).
Compliance Task Collection	*compliancetasks*	title, department, priority, dueDate, workStatus, assignedTo, activeWorkers (array of current workers), startedAt, completedAt, proofPhoto, proofNotes.
Daily Attendance Collection	*dailyattendances*	date (YYYY-MM-DD), records (array of employee punch logs, shift details, work duration, status), presentCount, absentCount.

5.2 Embedded Audit Trail Event Schema

Every change in ticket status or supervisor reassignment appends an immutable audit event to the complaint's auditTrail array, preserving full legal traceability for third-party compliance audits:

Event Attribute	Data Type	Description & Compliance Relevance
timestamp	Date	Exact UTC moment when the action occurred on the server.
action	String	Triggered action (e.g., CREATED, IN_PROGRESS, CAP_SUBMITTED, VERIFIED, REJECTED).
performedBy	Object	Captures { userId, employeeId, name, role } of the acting user.
notes / comment	String	Auditor findings, supervisor explanation, or rejection feedback.
previousStatus	String	State of the ticket prior to this event.
newStatus	String	New state achieved following successful validation.

API REFERENCE

6. RESTful API Specification & Endpoints

Comprehensive HTTP routing, authentication guards, and payloads

6.1 Endpoints Directory

Method	Route Endpoint	Auth Guard	Payload & Functional Description
POST	**/api/auth/login**	Public	Credentials { email, password } -> returns JWT Bearer token & user profile.
GET	**/api/auth/me**	Bearer	Returns currently authenticated user profile from token context.
GET	**/api/complaints**	Bearer	List all complaints with query filters (status, department, priority, deadline).
POST	**/api/complaints**	Auditor/Admin	Create NC ticket with multipart/form-data (includes beforePhoto).
PATCH	**/api/complaints/:id/in-progress**	Supervisor	Initiates countdown execution and sets status to In Progress.
POST	**/api/complaints/:id/submit-action**	Supervisor	Submits 9-field CAPA data and multipart afterPhoto proof.
POST	**/api/complaints/:id/verify**	Auditor/Admin	Submits verification verdict: approve (Verified/Closed) or reject.
GET	**/api/complaints/stats/kpi**	Bearer	Returns plant-wide KPI statistics, SLA breach counts, and risk radar.
GET	**/api/tasks/live-status**	Bearer	Returns live worker floor status, current jobs, and attendance summary.
GET	**/api/users**	Admin	Retrieves all plant users with dependency checks and status flags.
POST	**/api/users**	Admin	Provisions new employee credentials with role and department assignment.

6.2 Error Handling & Status Codes

All API responses follow consistent JSON envelopes. Success responses return { success: true, data: ... }. Errors return { success: false, message: ... } with standard HTTP semantics (400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found, 500 Internal Error).

ALGORITHMIC SPEC

7. Weighted Scoring Engine & SLA Calculations

Mathematical formulation of plant compliance and risk scoring

7.1 Compliance Scoring Algorithm

The scoring engine (server/utils/scoringEngine.js) executes an ISO-weighted formula across all audited checklist items and non-conformance defects. Severity categories carry mathematical penalties:

Defect Severity Level	Weight Penalty	Definition & Industrial Criteria
Critical Severity	10 Points	Issues halting production or causing severe quality/safety failures.
Major Severity	5 Points	Defects failing buyer specifications or causing customer rejects.
Minor Severity	2 Points	Workmanship skews, slight loose threads, or packaging blemishes.
Observation	1 Point	Floor housekeeping or preventive practice recommendations.

7.2 Mathematical Formula

The overall compliance percentage is calculated as: Score (%) = (Sum of (Item Status Score * Severity Weight)) / (Sum of (Maximum Possible Score * Severity Weight)) * 100, where Compliant = 100, Observation = 50, and Non-Compliant = 0. Items marked Not-Applicable (N/A) are mathematically excluded from both numerator and denominator.

7.3 Factory Risk Band Classifications

Risk Classification Band	Threshold Trigger Criteria	Mandatory Operational Response
CRITICAL RISK	Score < 60% OR >= 1 Uncontained Critical NC	Immediate plant manager intervention required; line stop triggered.
HIGH RISK	Score 60% - 74.9% OR Overdue CAP count > 3	Escalation to production manager; daily review audits mandated.
MEDIUM RISK	Score 75% - 89.9%	Standard operating risk; weekly containment audit required.
LOW RISK (EXCELLENT)	Score >= 90% AND Zero Critical NCs	Compliant with international AQL buyer standards; gold tier.

STANDARD PROCEDURES

8. Operational SOPs & Visual Defect Proof Workflows

Standard Operating Procedures for shop-floor defect rectification

8.1 Step-by-Step Defect Lifecycle SOP

Operational Phase	Mandatory Procedure & Protocol
Step 1: Floor Logging	Auditor identifies defect on floor, opens mobile portal, snaps Before-Photo, tags Department (e.g. Sewing Line 1), selects Category & Severity, and submits ticket. Timer activates.
Step 2: Line Supervisor Notification	Supervisor sees ticket badge turn Orange/Red on their Line View. Supervisor reviews defect photo, clicks "Start Work", and isolates affected garment bundles.
Step 3: Root Cause Isolation	Supervisor conducts 5-Why analysis with line mechanics/operators, rectifies machine setting or needle tension, and completes containment.
Step 4: CAPA Submission & Photo Upload	Supervisor uploads crisp After-Photo showing the rectified seam/garment, enters corrective and preventive actions, and clicks "Submit CAP Action".
Step 5: Physical Verification & Closure	Auditor receives notification, physically inspects next 20 pieces off the machine. If acceptable, clicks "Verify & Close". If defective, clicks "Reject for Rework" with feedback.

8.2 Photographic Evidence Standards

To guarantee audit validity, all photos must satisfy strict visual criteria: (1) High clarity under factory illumination, (2) Close-up macro view of the defect area, (3) Clean background, (4) Matching perspective between Before and After photos to enable side-by-side verification.

Automated Image Pre-Processing

The client-side image engine automatically normalizes all uploads to max 800px width/height and re-encodes as 85% quality JPEG. This guarantees fast mobile uploads while preventing PDF file bloat.

DEVOPS & DEPLOYMENT

9. Installation, Configuration & Deployment Guide

Setup instructions for development, Docker, and cloud production

9.1 System Prerequisites

• Node.js v18.0.0 or higher
• npm v9.0.0 or higher
• MongoDB v6.0+ instance (Local or MongoDB Atlas cluster)
• Modern web browser (Chrome, Edge, Safari, Firefox)

9.2 Environment Configuration (.env)

Create server/.env based on the template below:

```
PORT=5000
NODE_ENV=production
MONGODB_URI=mongodb+srv://<user>:<password>@cluster0.mongodb.net/garment_qms?retryWrites=true&w=majority
JWT_SECRET=super_secret_enterprise_jwt_key_2026
JWT_EXPIRES_IN=24h
CLIENT_URL=https://complainence-module.vercel.app
```

9.3 CLI Commands

Command	Action Executed
`npm run install:all`	Installs dependencies across root, server, and client workspaces simultaneously.
`npm run seed`	Populates initial factory departments, demo users, and test non-conformances into MongoDB.
`npm run dev`	Concurrently starts backend API (port 5000) and client Vite dev server (port 5173).
`npm run build`	Compiles production client bundle into client/dist ready for CDN/Vercel distribution.
`npm run docs:pdf`	Regenerates this comprehensive Project Documentation PDF.

MAINTENANCE & FAQ

10. Quality Auditing, Troubleshooting & FAQ

Common operational scenarios, error resolutions, and audit prep

10.1 Diagnostic Checklist & Troubleshooting

Observed Symptom	Root Cause & Resolution Procedure
Photos Missing in PDF	Verify that image URLs are accessible. The dual-mode PDF generator attempts Blob fetch with auth headers first, followed by Canvas decoding. Ensure CORS headers permit the host origin.
MongoDB Connection Timeout	Check that the IP address running the server is whitelisted on MongoDB Atlas. In offline mode, the system seamlessly falls back to in-memory mock store.
Token Expired / 401 Error	JWT tokens expire after 24 hours. The Axios response interceptor catches 401 responses and redirects users to login to maintain session security.
SLA Timer Inaccuracy	All timestamps are stored in ISO 8601 UTC format. Client components render elapsed time relative to the browser's local timezone (IST recommended).

10.2 Frequently Asked Questions (FAQ)

Q: Can additional factory departments be created dynamically?
A: Yes. Plant Administrators can add and manage departments (e.g. Embroidery, Washing, Dyeing) directly through the Administrative Portal.

Q: How does the system handle internet dropouts on the factory floor?
A: The client service utilizes local caching and resilient retry handlers. When offline, supervisors can still view cached tasks and queue updates.

Compliance Certification Sign-Off

This document confirms that the Garment QMS platform conforms to digital quality assurance standards, closed-loop defect rectification protocols, and ISO 9001:2015 traceability requirements.